



How does a geosynchronous earth-imaging satellite improve continuous environmental and disaster monitoring over India?

GS III - Science and Tech
Disaster Management

KNOWLEDGE POINTS

- **Fixed vantage point:** By matching Earth's rotation at 36,000 kilometres altitude, the satellite stays focused over one region. This eliminates gaps between passes, letting scientists observe the same landscape without waiting days for the spacecraft to circle back.
- **Real-time disaster tracking:** The spacecraft watches fast-evolving emergencies like flash floods, cyclones, and forest fires as they unfold. Continuous imaging helps emergency teams detect sudden environmental shifts instantly and respond before natural crises cause widespread community damage.
- **Advanced spectral scanning:** Instead of simple photography, specialized sensors record reflected light across multiple wavelengths. This technical capability lets researchers clearly distinguish farm crops, water bodies, and vegetation health, exposing subtle ecological changes across seasons.
- **Orbit trade-offs:** Low-earth satellites circle a few hundred kilometres up, capturing very sharp images but taking days to revisit any area. In contrast, higher geosynchronous satellites sacrifice pinpoint detail to deliver continuous regional views.
- **Preventing monitoring gaps:** Farmers sometimes burn crop stubble at odd hours to evade low-orbit satellites that pass only at fixed times. Constant observation ensures continuous monitoring, making it impossible for polluters to hide smoke emissions.

CONCEPT UNFOLD

Continuous observation of the same area improves monitoring →



Changes can be detected quickly

When a satellite keeps watching the same region, sudden events such as floods, fires or cyclones can be noticed without waiting for another satellite pass.



Faster detection supports faster response

When authorities receive information quickly, they get more time to issue warnings, plan evacuations and send help before a disaster becomes more dangerous.



Repeated observations improve situation tracking

Continuous images allow authorities to compare conditions over time, helping them track how a cyclone, flood or wildfire is developing and adjust their response accordingly.

KEY TAKEAWAY

By operating continuously from geosynchronous orbit, EOS-05 replaces intermittent satellite passes with uninterrupted spectral monitoring, giving disaster responders and environmental managers instant views of rapid ground changes across India.



How has the Strait of Hormuz trapped the United States in an unwinnable regional conflict?

GS II - Policies
Developed and Developing Countries

KNOWLEDGE POINTS

- **Strategic battlefield shift:** Tehran turned the narrow waterway into the main arena of war. This move completely shifted focus away from America's original goals of destroying nuclear sites or changing Iran's government, forcing Washington into a frustrating tactical maritime battle.
- **Asymmetric deterrence power:** Iran repeatedly targets commercial ships and military vessels using geography to its advantage. Even heavy economic sanctions and naval blockades have failed to break Iran's chokehold, leaving America unable to secure the shipping lane or claim decisive victory.
- **Regional base vulnerability:** Whenever American forces strike Iranian targets or tankers, Iran retaliates directly against American military bases in Kuwait, Bahrain, and Jordan. This pushback raises the human and material cost of the war, trapping American forces in endless counter-strikes.
- **Mowing the lawn:** Instead of achieving a decisive political breakthrough, Washington engages in repetitive strikes simply to blunt Iranian capabilities. This attritional cycle never eliminates underlying threats, as each wave of attacks provokes bolder Iranian responses.
- **Failed peace diplomacy:** The collapse of the June 17 Islamabad agreement removed formal diplomatic guardrails. Without open communication channels or agreed ceasefire terms, minor tactical clashes at sea quickly escalate into wider regional conflict.

CONCEPT UNFOLD

Heavy dependence on a maritime chokepoint increases vulnerability →



A local disruption can affect many countries

When large amounts of oil and gas pass through one narrow route, war or blockage there can interrupt energy supplies far beyond the conflict area.



Energy costs can rise across the economy

Reduced supply can make oil and gas more expensive. Higher fuel costs then increase transport and production expenses, making many everyday goods costlier.



Countries may seek alternative supply routes

Repeated disruption risks can encourage governments and companies to diversify energy suppliers, transport routes and strategic reserves to reduce dependence on a single chokepoint.

KEY TAKEAWAY

By turning the Strait of Hormuz into an asymmetric battleground, Iran deflected Washington's primary war objectives and trapped American forces in an escalating, open-ended attritional conflict without any clear exit strategy.